

RESEARCH ARTICLE

A Hybrid Chatbot Model for Enhancing Administrative Support in Education: Comparative Analysis, Integration, and Optimization

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ABSTRACT The increasing number of students and the limited administrative resources in educational institutions have highlighted inefficiencies in traditional communication methods. Although chatbots offer potential solutions, current implementations are constrained by their architectures: rule-based chatbots lack flexibility, retrieval-based systems depend heavily on predefined datasets, and generative models risk inaccuracies. This study introduces a hybrid chatbot model designed to overcome these challenges by combining rule-based, retrieval-based, and generative approaches. The proposed model employs a Multinomial Naive Bayes classifier to intelligently route user queries to the most appropriate module, ensuring optimized performance across diverse query types. A domain-specific dataset from Sakarya University was developed, featuring 2,253 question-answer pairs categorized into seven administrative domains. Experimental results demonstrate that the hybrid chatbot model outperforms standalone approaches, achieving 97.57% accuracy and improved user satisfaction. Moreover, the model's flexibility allows seamless adaptation to other sectors, such as healthcare and e-commerce. By significantly reducing administrative workload and enhancing engagement, the proposed hybrid chatbot provides an effective framework for scalable, robust, and adaptable conversational agents in educational settings and beyond.

INDEX TERMS Rule-based, retrieval-based, generative-based, hybrid model, chatbots, educational administration.

I. INTRODUCTION

Efficient communication among students, faculty, and administrators is fundamental to the smooth functioning of educational institutions. However, the rapid growth in student populations and the increasing complexity of administrative tasks have exposed limitations in traditional communication methods, such as in-person interactions, emails, and phone calls [1]. These challenges often result in delays, inefficiencies, and a reduced quality of service, creating barriers to academic success [2], [3].

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Chatbots, as conversational agents, present a promising solution for addressing these communication bottlenecks [4]. By providing 24/7 accessibility, chatbots can significantly reduce the workload of administrative staff while improving response times and user satisfaction [5]. Despite these advantages, existing chatbot technologies exhibit notable limitations. Rule-based chatbots are rigid and confined to predefined scripts, making them unsuitable for complex or open-ended queries [6]. Retrieval-based chatbots rely heavily on pre-existing datasets, which limits their adaptability to ambiguous or nuanced inquiries [7]. Meanwhile, generative chatbots, powered by advanced language models, offer flexibility but often produce inaccurate or contextually irrelevant

responses, an issue particularly concerning in educational settings where reliability and precision are paramount [8].

To address these limitations, this study proposes a hybrid chatbot model that integrates the strengths of rule-based, retrieval-based, and generative approaches. The model leverages a Multinomial Naive Bayes (MNB) classifier to intelligently route queries to the most suitable chatbot module, ensuring optimized performance across diverse query types. This hybrid framework aims to provide reliable, scalable, and context-aware support for administrative tasks, tailored specifically to the needs of educational institutions.

The contributions of this research are threefold:

1. A comprehensive evaluation of rule-based, retrieval-based, and generative chatbot approaches in handling educational administrative queries.
2. The development and validation of a hybrid chatbot model, trained on a domain-specific dataset comprising 2,253 categorized question-answer pairs from Sakarya University.
3. Demonstration of the model's scalability and adaptability for other domains, including healthcare and e-commerce, through its modular and domain-agnostic design.

The remainder of this article is organized as follows: Section II presents a review of related work, highlighting the limitations of existing chatbot technologies. Section III details the methodology, including dataset creation, model development, and the integration of the hybrid framework. Section IV discusses the results, and Section V concludes with insights and future research directions.

II. LITERATURE REVIEW

The growing interest in chatbots within educational settings has spurred research into their potential for enhancing student learning experiences and streamlining administrative tasks [1]. Chatbots offer 24/7 accessibility, foster increased student engagement, and reduce administrative workload [9], [10]. The existing literature highlights two primary areas of application: information provision and administrative support.

A. APPLICATIONS OF CHATBOTS IN EDUCATIONAL ADMINISTRATION

Chatbots in education are widely used to address FAQs and provide general information on topics such as academic schedules, admissions, library services, tuition fees, and campus life [11], [12], [13], [14], [15], [16]. They act as 24/7 virtual assistants, ensuring quick and accurate responses to student queries, reducing waiting times, and improving accessibility. Additionally, chatbots can personalize information based on individual user needs, making their interactions more relevant and effective [17], [18]. Beyond providing information, chatbots automate key administrative tasks, such as guiding prospective students through the admissions process, assisting with course registration, and handling fee payment process [19], [20], [21]. By streamlining these

processes, chatbots reduce the workload for administrative staff, minimize errors, and ensure greater efficiency in routine operations [22], [23]. The use of chatbots offers several advantages. They enhance operational efficiency, reduce labor costs, and allow staff to focus on more complex tasks [24], [25]. Furthermore, their ability to provide consistent, instant support improves user experience and boosts student satisfaction. By operating round the clock, chatbots create a more inclusive and accessible support system for all stakeholders [26], [27], [28].

Despite these advancements, the effectiveness of chatbots depends largely on their architecture. The following sections examine the strengths and limitations of rule-based, retrieval-based, and generative chatbot models as reported in the literature.

B. CHATBOT ARCHITECTURES

1) RULE-BASED CHATBOTS

Rule-based chatbots rely on predefined rules and patterns for matching user inputs to specific responses. Their simplicity and predictability make them suitable for addressing well-defined queries, such as tuition deadlines or course schedules [3], [10], [29]. However, their reliance on fixed patterns makes them inflexible for handling open-ended or ambiguous questions [5]. Development tools like Artificial Intelligence Markup Language (AIML) have been widely used for creating rule-based chatbots [30], [31], [32].

2) RETRIEVAL-BASED CHATBOTS

Retrieval-based chatbots match user inputs to the most relevant responses within a predefined dataset using machine learning and similarity algorithms. Frameworks like DialogFlow, RASA, and Chatterbot have been employed in educational chatbot implementations [21], [33], [34], [35], [36], [37], [38]. While retrieval-based models excel in providing accurate responses for specific queries, their reliance on dataset quality and scope limits their ability to handle novel or nuanced questions [5].

3) GENERATIVE CHATBOTS

Generative chatbots, powered by neural networks, generate responses dynamically rather than relying on pre-written answers. Technologies such as Recurrent Neural Networks (RNNs) and transformer-based architectures like GPT-3 and Pegasus have shown promise in improving chatbot flexibility and adaptability [39], [40]. Generative models are particularly valuable in educational settings for open-ended tasks, such as providing personalized explanations or creative prompts [41], [42]. However, these models often struggle with inaccuracies, hallucinations, and contextual misunderstandings, making them less reliable for critical administrative tasks [8], [43].

C. COMPARATIVE ANALYSES AND HYBRID APPROACHES

Comparative studies have evaluated the strengths and weaknesses of these chatbot architectures. For example, a study

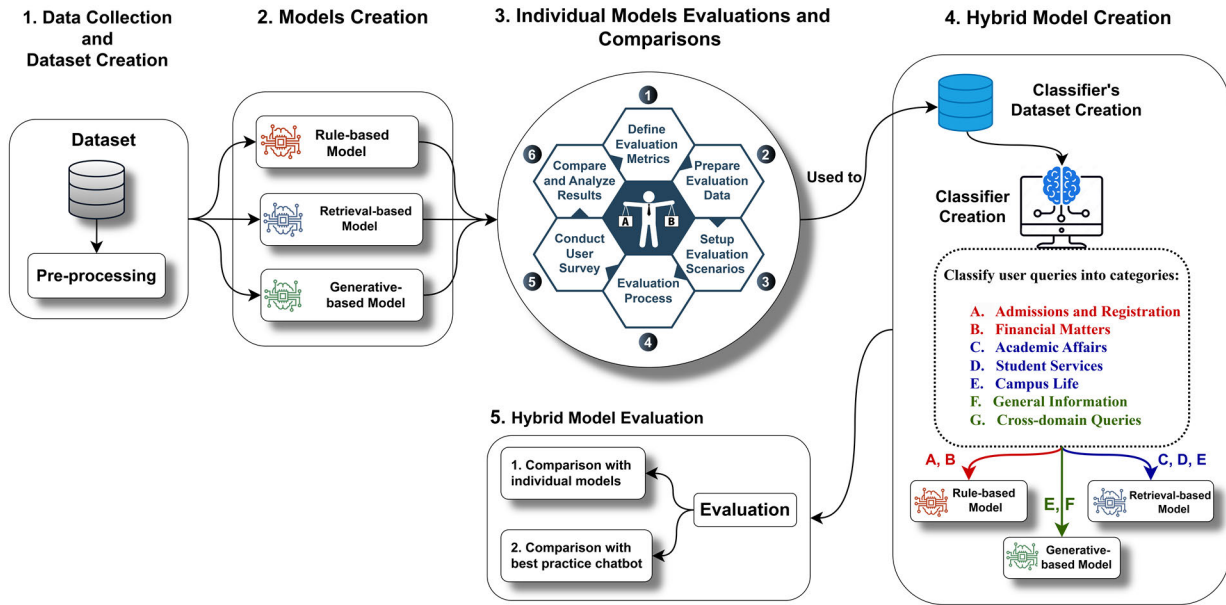


FIGURE 1. Workflow diagram.

comparing AIML-based rule-based models and sequence-to-sequence generative models found that while rule-based chatbots excel in task completion, they often fall short in user satisfaction, especially for open-ended queries [30]. Similarly, a study on retrieval-based models highlighted their effectiveness in structured domains but noted their limitations in dynamic and ambiguous scenarios [1].

Hybrid chatbot models, which combine the strengths of multiple architectures, have emerged as a promising solution to overcome these limitations. For instance, integrating retrieval-based and generative models has been shown to improve adaptability without sacrificing reliability [44], [45]. However, the literature lacks a comprehensive evaluation of hybrid systems in the context of educational administration, particularly in handling diverse and complex query types.

D. RESEARCH GAP

While significant progress has been made in developing chatbot systems for educational purposes, critical gaps remain:

Scalability and Generalizability: Many studies focus on datasets tailored to specific institutions, limiting the applicability of their findings to broader contexts [3], [46].

Ethical and Privacy Considerations: There is limited discussion on how chatbots address data privacy and ethical concerns, particularly in handling sensitive student information [17], [18].

Performance in Complex Scenarios: Existing models often underperform in managing cross-domain queries or providing personalized assistance [8], [43].

E. SUMMARY

The existing body of work underscores the potential of chatbots to transform educational administration by automating

routine tasks and improving accessibility. However, the limitations of individual architectures highlight the need for innovative approaches that integrate their strengths. This study addresses these gaps by proposing a hybrid chatbot model specifically designed for educational administrative support. By leveraging rule-based, retrieval-based, and generative components, the proposed hybrid model aims to deliver superior accuracy, reliability, and scalability, setting a new benchmark for chatbot performance in educational contexts.

III. METHODOLOGY

This section outlines the development and evaluation process for the proposed hybrid chatbot model. The methodology, illustrated in Figure 1, consists of five key phases: data collection and dataset creation, development of individual models, evaluation and comparison of models, proposed hybrid model, and finally, evaluation processes.

A. DATA COLLECTION AND DATASET CREATION

The primary objective of the proposed system is to develop a chatbot framework with generic features, enabling it to adapt to the administrative requirements of various educational institutions. Although the system is designed to support diverse use cases and institutional needs, Sakarya University was selected as a case study to demonstrate its feasibility and effectiveness. Since no existing datasets were specifically tailored to Sakarya University, a new dataset was created from scratch to support the system's development. This dataset consists of a collection of questions and corresponding answers stored in a structured text file format. To ensure clarity and consistency, a standardized notation system was adopted: questions are prefixed with "Q:" and answers with "A:".

The data was sourced from multiple channels to ensure comprehensive coverage. Sources included Sakarya University's FAQ pages, student discussions in social media groups, and information available on the university's official websites. This diverse data-gathering approach allowed the dataset to capture a wide range of inquiries commonly encountered by students. Additionally, valuable feedback from students was integrated into the dataset, enhancing its relevance and applicability.

To prepare and organize the collected information, several data pre-processing steps were implemented:

- **Text Cleaning:** involved removing unnecessary elements such as whitespace, special characters, and irrelevant symbols such as hashtags from social media, conducting spell-checking to correct typographical errors, and eliminating redundant entries to ensure a balanced dataset.
- **Structuring and Parsing:** ensured that all questions and answers were correctly paired using “Q:” and “A:” markers, resolved missing pairs, and split lines systematically to improve dataset organization.
- **Handling Diverse Sources:** standardized terminology by replacing variations like “Uni” with “University,” harmonized phrasing across data from different sources such as social media and student feedback, and filtering out irrelevant entries to maintain focus.
- **Ethical and Privacy Considerations:** ensured compliance with data protection standards by removing sensitive information and anonymizing personal identifiers to create an ethically sound dataset.

The final dataset comprises 2,253 lines of conversation, categorized into seven distinct domains: Admissions and Registrations, Academic Affairs, Financial Matters, Student Services, Campus Life, General Information, and Cross-domain Queries. This categorization ensures broad coverage of topics, addressing the needs of prospective, current, and international students at Sakarya University. The dataset's structured and comprehensive design provides a strong foundation for developing and evaluating the chatbot system while ensuring its adaptability to various administrative contexts.

B. INDIVIDUAL MODELS DEVELOPMENT

Three chatbot architectures rule-based, retrieval-based, and generative were developed to address diverse types of queries encountered in educational administrative tasks.

1) RULE-BASED CHATBOT

The rule-based chatbot was developed using AIML (Artificial Intelligence Markup Language), selected for its ease of use, adaptability, and natural language processing capabilities [46]. AIML employs XML-based syntax to define conversational rules, enabling the chatbot to map user inputs to predefined patterns and generate appropriate responses. Each conversational rule in AIML is structured as a category, the fundamental unit of AIML. A category comprises two key elements: a pattern, which specifies the input query or phrase the chatbot is designed to recognize, and a template,

which defines the corresponding response. Figure 2 provides an example of an AIML rule, demonstrating how these components work together to facilitate dynamic and structured interactions.

```
<aiml>
  <category>
    <pattern>HELLO</pattern>
    <template>Hello, how can I assist you today?</template>
  </category>
  <category>
    <pattern>WHAT IS YOUR NAME</pattern>
    <template>My name is Sakarya Bot.</template>
  </category>
</aiml>
```

FIGURE 2. A sample of AIML rule.

AIML utilizes a pattern-matching mechanism to compare predefined input patterns with user queries. These patterns can incorporate wildcards such as *, which captures variable-length segments of a sentence, and _, which matches a single word. This feature provides flexibility, allowing the chatbot to process a wide range of user inputs using a single rule. Figure 3 illustrates an example of pattern matching with wildcards, showcasing how AIML handles diverse conversational inputs effectively.

```
<category>
  <pattern>WHAT IS *</pattern>
  <template>I'm sorry, I don't know the answer to that question.</template>
</category>
```

FIGURE 3. Pattern matching and wildcards rule.

The rule shown in Figure 3 matches any query starting with “What is” and delivers a generic response. Beyond basic pattern matching, AIML offers advanced features such as recursive calls, variables, and conditional logic, which enhances its ability to generate dynamic and context-aware responses. Recursive calls enable templates to reference other categories, supporting the creation of complex and interconnected dialogues. Variables allow the chatbot to store and retrieve user-specific information, such as names or preferences, to personalize interactions. Conditional logic adds decision-making capabilities, enabling responses to adapt based on contextual information. Figure 4 illustrates an example of an AIML variable, where the name variable captures user-provided names and allows the chatbot to remember and use them throughout the conversation.

The chatbot was developed and deployed using the Pandorabots platform [47], a cloud-based environment known for its user-friendly interface and robust support for AIML-based chatbot creation. The platform streamlines the development process by offering tools such as an integrated AIML editor for managing knowledge bases, a testing console for validating responses, and a natural language processing engine optimized for efficient pattern matching. Additionally, Pandorabots provides scalable hosting, ensuring chatbot

```

<category>
  <pattern>MY NAME IS *(</pattern>
  <template>
    <set name="username"><star/></set>
    Hello, <get name="username"/>! How can I assist you today?
  </template>
</category>

```

FIGURE 4. AIML variable example.

responsiveness even during periods of high traffic. One of its standout features is a RESTful API, enabling seamless integration of chatbots into diverse applications, including websites, mobile apps, and messaging platforms, to facilitate multi-channel interaction.

The development process with AIML and Pandorabots followed a structured workflow. First, the chatbot's knowledge base was established by defining 637 AIML rules, each addressing a specific administrative query or task. These rules were derived from the dataset created and uploaded to the Pandorabots platform, where they were compiled into a structured knowledge base. The chatbot's functionality was rigorously tested and refined using the platform's testing console, ensuring accurate input pattern matching and context-appropriate responses.

Once refined, the chatbot was deployed via the Pandorabots API, allowing integration with platforms such as a university website or mobile application to support student queries. Post-deployment, the chatbot's performance was monitored continuously, with periodic updates to its knowledge base to address emerging queries and enhance response accuracy.

The prototype was tested by 113 participants, comprising both undergraduate and postgraduate students. Interaction logs were analyzed to evaluate response accuracy and user satisfaction, resulting in iterative improvements to the chatbot's functionality and overall effectiveness. This testing phase ensured that the chatbot met the needs of its intended audience and operated as a reliable tool for administrative support.

2) RETRIEVAL-BASED CHATBOT

The development of the retrieval-based chatbot followed a structured approach to knowledge management and response retrieval, leveraging a JSON-based framework as the foundation for its knowledge base. The JSON structure was tailored to Sakarya University and organized into the same primary categories as the dataset described earlier. Each entry in the JSON file is composed of four main components: intents, questions, responses, and entities.

- Intents represent the specific categories of user inquiries, providing a clear organizational layer for mapping queries.
- Questions are arrays of user input examples associated with each intent, helping the chatbot identify and classify user queries.

- Responses include predefined answers relevant to each intent, designed to address user needs effectively.
- Entities capture key contextual details extracted from user inputs, such as course names, dates, or other attributes, enabling the chatbot to deliver personalized and context-aware responses.

This hierarchical organization ensures efficient mapping of user queries to their relevant domains, accurate detail extraction, and effective response generation. To enhance the chatbot's accuracy and efficiency, comprehensive data preprocessing steps were applied to the text data. These steps included:

- Tokenization: Breaking down text into individual units or words to simplify further analysis.
- Stopword Removal: Filtering out common words (e.g., "the," "and") that contribute little semantic value.
- Stemming: Using the Porter Stemmer algorithm to reduce words to their root forms (e.g., "running" to "run"), ensuring consistent text representation across queries and responses.

After preprocessing, a clean and standardized dataset was created by pairing processed user queries (patterns) with randomly selected pre-processed responses from the JSON structure. This approach ensured that the dataset was consistent, ready for analysis, and optimized for response matching.

To evaluate the relevance of user queries, the chatbot utilized Term Frequency-Inverse Document Frequency (TF-IDF), a statistical measure that determines the importance of words within a document relative to their frequency in the dataset [48]. This methodology enabled the chatbot to prioritize meaningful terms and enhance the accuracy of its query-to-response matching. The TF-IDF score for a term t in a query q is computed as:

$$TF - IDF(t, q, D) = TF(t, q) \times IDF(t, D) \quad (1)$$

where

- Term Frequency (TF):

$$TF(t, q) = \frac{f_{t,q}}{\sum_{t' \in d} f_{t',q}} \quad (2)$$

$f_{t,q}$ is the frequency of term t in query q , and $\sum_{t' \in d} f_{t',q}$ is the total number of terms in query d .

- Inverse Document Frequency (IDF):

$$IDF(t, D) = \log\left(\frac{N}{1 + DF(t)}\right) \quad (3)$$

N is the total number of predefined questions in the dataset D , and $DF(t)$ is the number of predefined questions containing the term t . This ensures common terms like "the" or "and" are given less weight, while domain-specific terms such as "admissions", "tuition" are emphasized.

Once the TF-IDF vectors for both the user query and the predefined questions are generated, cosine similarity is used to measure their semantic similarity:

$$\text{cosine_similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} \quad (4)$$

where:

- A and B are the TF-IDF vectors of the query and a predefined question.
- $A \cdot B$ is the dot product of the vectors.
- $\|A\|$ $\|B\|$ are the magnitudes of A and B, respectively.

The response selection process of the retrieval-based chatbot relies on cosine similarity to determine the closest match to a user query. Among the predefined questions in the dataset, the one with the highest cosine similarity score is identified as the best match, and the corresponding response is retrieved.

To enhance contextual understanding, the chatbot employs an entity extraction functionality, implemented through a user-defined function named `extract_entities`. This function processes user inputs in three key steps:

- **Tokenization:** Splits the user input into individual words or tokens.
- **Pattern Matching:** Compares the input against predefined patterns associated with specific intents to identify potential matches.
- **Entity Extraction:** Uses regular expressions to extract specific elements from the matched patterns, such as names, dates, or keywords.

This approach allows the chatbot to incorporate user-specific details into its responses, ensuring they are both accurate and contextually relevant. The response generation workflow is designed to deliver precise and efficient answers:

- **Input Preprocessing:** The user query undergoes tokenization, stopword removal, and stemming to standardize its format.
- **Similarity Calculation:** The system calculates the cosine similarity between the processed input and pre-processed dataset questions to identify the most relevant intent.
- **Entity Extraction:** Using the `extract_entities` function, the chatbot extracts key details from the user query related to the identified intent.
- **Response Retrieval:** The chatbot generates the final response by combining the identified intent with the extracted entities to provide a customized and context-aware answer.

This multi-step process ensures that the chatbot delivers responses that are not only accurate but also tailored to the specific context of each user query.

3) GENERATIVE-BASED CHATBOT

The development of the generative-based chatbot incorporated advanced technological components to enhance its adaptability and effectiveness in addressing diverse queries. The process began with the selection of a pre-trained transformer model from the Hugging Face library, renowned for its cutting-edge natural language processing capabilities. The Pegasus architecture was chosen as the base model due to its excellence in sequence-to-sequence tasks, such as text generation and summarization. This pre-trained model served

as a foundation, enabling fine-tuning to meet the chatbot's specific requirements.

The training dataset, derived from the Q&A dataset created for this research, was tailored to the chatbot's application domain and underwent extensive pre-processing. The pre-processing phase included:

- **Tokenization:** Dividing text into smaller, meaningful units for further processing.
- **Encoding:** Converting tokens into numerical representations suitable for input into the transformer model.
- **Padding and Truncation:** Ensuring all input sequences were of consistent length, a critical requirement for efficient model training.

This processed dataset was instrumental in fine-tuning the generative model, equipping it to produce contextually accurate and high-quality responses. The fine-tuning process focused on optimizing the model's performance with the custom dataset. The Adam optimizer was employed for its adaptive learning rate capabilities, facilitating efficient convergence. Training occurred over 40 epochs, with each epoch representing a complete pass through the dataset. A loss function quantified the model's prediction errors compared to the expected outcomes, and the model weights were iteratively adjusted to minimize these losses. This optimization process refined the model's ability to generate coherent, relevant, and grammatically accurate responses.

Upon completing fine-tuning, the model underwent rigorous evaluation to assess its performance across three key criteria:

- **Coherence:** Ensuring responses were logically structured and contextually appropriate.
- **Relevance:** Measuring the alignment of generated responses with user queries.
- **Grammatical Accuracy:** Confirming high linguistic quality in generated outputs.

The model was tested using various query scenarios, including open-ended and complex questions, to validate its robustness and adaptability. Its dynamic response generation capabilities proved particularly effective for handling ambiguous or creative inquiries, highlighting the chatbot's versatility and making it well-suited for a wide range of user interactions.

C. MODELS' EVALUATION AND COMPARISON

This section presents an in-depth analysis of the three chatbot models developed earlier, examining their strengths, weaknesses, and unique performance characteristics. Additionally, the feasibility of a hybrid model that combines the most advantageous features of all three approaches is explored. Figure 5 illustrates the six-step lifecycle of the comparison process, providing an overview of the evaluation methodology.

1) DEFINE EVALUATION METRICS

The performance of the chatbot models was assessed using a combination of core metrics and supplementary metrics:

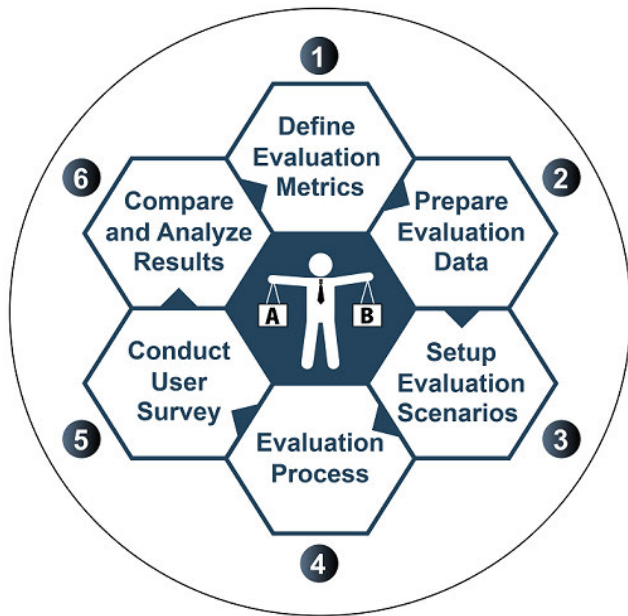


FIGURE 5. Models' evaluation and comparison life cycle.

- Core Metrics included accuracy (the proportion of inquiries correctly and relevantly answered), precision (the percentage of responses that were topical and directly related to the user's query), recall (the proportion of relevant questions accurately addressed), and F1 score (a harmonic mean of precision and recall, offering a balanced performance measure).
- Supplementary Metrics included user satisfaction and engagement (assessed through participant surveys to capture qualitative feedback) and flexibility (evaluated based on the chatbot's ability to adapt to new information and handle unexpected or complex queries).

2) PREPARE EVALUATION DATA

A realistic and diverse dataset of 300 representative queries was compiled for evaluation. These queries were selected from Sakarya University's FAQ pages and represented approximately 15% of the training dataset size. The dataset encompassed a wide range of administrative tasks, capturing real-world user inquiries across different domains. This curated dataset provided a robust foundation for assessing the models' capabilities in meeting the complex and dynamic needs of users.

3) SETUP EVALUATION SCENARIOS

Three distinct scenarios were defined to simulate the chatbot's functionality across various use cases:

- Scenario 1: Queries related to admissions, academic programs, and general information for prospective students.
- Scenario 2: Inquiries about registration, tuition fees, courses, and campus facilities for enrolled students.
- Scenario 3: Queries addressing admission processes, tuition fees, weather conditions, language requirements, and accommodation for foreign students.

These scenarios were designed to reflect the diverse needs of Sakarya University's stakeholders, enabling a comprehensive assessment of the models across multiple user categories.

4) EVALUATION PROCESS

Each chatbot model was evaluated based on its specific design and intended purpose:

- Rule-Based Chatbot was assessed for its ability to match user queries to predefined rules, with metrics such as rule coverage and matching accuracy used to measure performance.
- Retrieval-Based Chatbot was evaluated for retrieving the most relevant responses from the predefined knowledge base, using metrics such as accuracy, precision, recall, and F1 score.
- Generative-Based Chatbot was assessed for the quality of dynamically generated responses, considering factors such as coherence, grammatical accuracy, and relevance to user queries.

5) CONDUCT USER SURVEY

To gain insights into user satisfaction, a survey was conducted with 267 undergraduate students, who completed a 10-question questionnaire. Participants rated aspects such as:

- Response accuracy
- Usefulness of the chatbot
- Overall user experience

The feedback provided valuable insights into the usability and effectiveness of each chatbot model, as well as areas requiring improvement. This ensured that the evaluation process incorporated both technical performance metrics and user-centric perspectives.

6) COMPARE AND ANALYZE RESULTS

The final step involved a detailed comparison of the models based on the evaluation metrics and user feedback:

- Rule-Based Chatbot demonstrated strengths in handling predefined rules and structured queries but struggled with open-ended or ambiguous queries, limiting its versatility.
- Retrieval-Based Chatbot excelled at retrieving precise answers from the knowledge base when the dataset was well-curated but showed weaknesses due to its performance being heavily dependent on the quality and scope of the dataset.
- Generative-Based Chatbot produced dynamic responses and handled open-ended queries effectively but occasionally exhibited grammatical inaccuracies or coherence issues in complex scenarios.

This comparative analysis revealed the strengths and limitations of each approach, providing key insights into their performance characteristics. The rule-based chatbot demonstrated reliability for structured tasks, while the retrieval-based model excelled in retrieving precise responses. The generative-based model stood out for its

adaptability but required further refinement to address linguistic inconsistencies. These findings informed the development of a hybrid chatbot system that integrates the strengths of all three approaches, offering a versatile solution capable of meeting a broader range of user needs.

D. DEVELOPED HYBRID MODEL

Building upon the individual strengths of the rule-based, retrieval-based, and generative-based models, this section introduces the proposed hybrid chatbot system, which integrates these models into a unified and cohesive framework. The integration process centres around two primary components: the classifier and the main program loop.

The classifier, implemented using the Multinomial Naive Bayes (MNB) algorithm as described in Section C.1, is critical for efficiently routing user queries to the appropriate model. By categorizing queries into predefined labels based on their content and context, the classifier ensures that each query is directed to the model best suited for generating an accurate and relevant response. For instance:

- Queries requiring deterministic answers are routed to the rule-based model.
- Inquiries demanding specific matches from a structured knowledge base are directed to the retrieval-based model.
- Open-ended or complex questions are forwarded to the generative-based model for dynamic response generation.

The main program loop manages the interaction flow between users and the system, ensuring seamless communication. It acts as the central control mechanism, coordinating between the classifier and the individual chatbot models. This loop involves the following steps:

- Accepts and pre-processes user queries.
- Passes the processed queries to the classifier for categorization.
- Routes the query to the selected chatbot model based on the classifier's output.
- Retrieves the response from the chosen model and delivers it to the user.

This tightly integrated structure enables the hybrid chatbot system to leverage the unique strengths of each model while minimizing their individual limitations, resulting in a versatile and robust solution capable of addressing a wide range of user queries effectively.

The classifier and the main program loop are described further in the next two sections.

1) CLASSIFIER SELECTION, IMPLEMENTATION, AND EVALUATION

Based on the evaluation and comparison of the models, the strengths of each individual approach were identified, informing the creation of a specialized dataset for training the classifier. A sample of this dataset is presented in Table 1. It includes seven output labels, each with one hundred samples, ensuring comprehensive coverage of diverse

administrative topics relevant to Sakarya University. The classifier plays a critical role in balancing efficiency and performance by categorizing user queries into predefined labels, such as Admissions, Financial Matters, and General Information. This categorization ensures that each query is routed to the most suitable chatbot model for processing.

Four classifiers: Multinomial Naive Bayes (MNB), Support Vector Machines (SVM), Random Forest (RF), and Neural Networks (NN) were evaluated using this dataset. The evaluation considered computational efficiency alongside performance metrics. As detailed in Table 9 (discussed in the subsequent section), MNB emerged as the most practical choice for real-time deployment in educational administrative systems. Known for its effectiveness in sentiment analysis and multi-class categorization scenarios, MNB is widely used for categorical classification tasks [49], [50], [51]. As a probabilistic machine learning algorithm, MNB assumes that the occurrence of a word in a document is independent of the occurrence of other words, a property well-suited for this application [52].

The implementation of the MNB classifier is outlined in Figure 6. The process begins with the initialization of inputs, including the training dataset D , which consists of labelled examples in the form of (Query, Category) pairs. For instance:

- The query, "How do I register for classes?" is categorized under Admissions and Registration.
- The query, "What is the deadline to pay fees?" falls under Financial Matters.

The predefined categories C include: Admissions and Registration, Financial Matters, Academic Affairs, Student Services, Campus Life, General Information and Cross-domain Queries.

Additionally, a vocabulary V is constructed during this phase, encompassing all unique words extracted from the queries in the training dataset. These inputs serve as the foundation for training the classifier.

The next step involves pre-processing the text data to ensure consistency and focus on meaningful features. Each query is tokenized into individual words, and normalization techniques are applied:

- Lowercasing: Converts all text to lowercase for uniformity.
- Stopword Removal: Eliminates common words such as "and", "the", and "is" that contribute little semantic value.
- Stemming/Lemmatization: Reduces words to their root forms, for example, "registering" becomes "register".

Following pre-processing, the vocabulary V is finalized, containing all unique terms from the dataset. This vocabulary serves as the feature set, enabling the conversion of text queries into numerical representations for training and prediction. The MNB classifier leverages this structured input to efficiently categorize queries, ensuring accurate and scalable query handling within the chatbot system.

The training process involves calculating two sets of probabilities: prior probabilities $P(c)$ and likelihood probabilities

$P(w|c)$. Prior probabilities represent the likelihood of a query belonging to a particular category c and are computed as:

$$P(c) = \frac{N_c}{N}, \quad (5)$$

where N_c is the number of queries in category c , and N is the total number of queries in the dataset.

Likelihood probabilities represent the probability of a word w appearing in queries of category c . To calculate these probabilities, Laplace smoothing is applied to handle cases where a word does not appear in a particular category. The likelihood probabilities are computed as:

$$P(w|c) = \frac{\text{count}(w, c) + 1}{N_c^{\text{words}} + |V|} \quad (6)$$

where $\text{count}(w, c)$ is the frequency of word w in queries belonging to category c , N_c^{words} is the total count of all words in queries of category c , and $|V|$ is the size of the vocabulary.

To predict the category of a new query q , the query is first tokenized and preprocessed using the same methods applied during the training phase. The system then calculates the log-posterior probability for each category c . This is done using the formula:

$$\log P(c|q) = \log P(c) + \sum_{w \in q} \log P(w|c) \quad (7)$$

where the summation iterates over all words w in the query q , without needing an explicit upper bound, as it is defined by the words in the query itself. This formula combines the prior probability $P(c)$ with the likelihood probabilities $P(w|c)$ for all words in the query. The query is then assigned to the category with the highest log-posterior probability, expressed as:

$$\text{Predicted Category} = \text{argmax}_c(\log P(c|q)) \quad (8)$$

Finally, the system returns the predicted category for the input query, completing the classification process. This approach ensures that student queries are accurately categorized into the predefined set of query types, enabling efficient handling and response generation for a wide range of inquiries.

2) THE MAIN PROGRAM LOOP

The hybrid chatbot model operates through a well-structured process designed to classify and respond to user queries with accuracy and relevance. The process begins when the system initializes and is ready to process input. The primary input is the UserQuery, a natural language question or request submitted through the system interface. Examples of such queries include, “What are the registration deadlines?” or “How do I pay my tuition fees?” The system processes this input and generates a tailored Response, which is displayed to the user, ensuring accuracy and contextual relevance.

At the core of the model is the QueryProcessing Function, which classifies each query into one of several predefined categories using a trained classifier. These categories include:

TABLE 1. Sample of classifier dataset.

Query	Category
What are the exchange programs available at Sakarya University?	Academic Affairs
Are the library working hours changing during the exam period?	Student Services
Which students must pay the tuition fees?	Financial Matters
Does Sakarya experience any extreme weather conditions such as earthquakes?	General Information
What are the different types of dormitories available at Sakarya University?	Campus Life

Admissions and Registration, Financial Matters, Academic Affairs, Student Services, Campus Life, General Information, and Cross-domain Queries.

Classification determines which model is best suited to process the query:

- For queries in categories like Admissions and Registration (A) or Financial Matters (B), the system employs the Rule-Based Model, which handles structured queries with predefined responses.
- Queries related to Academic Affairs (C), Student Services (D), or Campus Life (E) are routed to the Retrieval-Based Model, which selects the most relevant response from a predefined knowledge base.
- Complex or open-ended queries categorized as General Information (F) or Cross-domain (G) Queries are directed to the Generative-Based Model, which dynamically generates responses.

For unclassified queries that do not fit into any predefined category, the system generates a fallback response, such as, “Sorry, I cannot handle this query at the moment.” This ensures that users always receive a reply, even if their query falls outside the existing framework.

Once the query is processed by the appropriate model, the generated response is passed back to the main program loop, which manages the user interaction cycle. The program loop takes the response from the QueryProcessing Function and delivers it to the user via the system interface, marking the completion of the interaction for that query. The process concludes once the response has been successfully displayed.

Figure 7 illustrates the structural components of the hybrid model, while Figure 8 provides the pseudocode for its process flow, outlining the systematic approach to handling user queries and generating contextually appropriate responses.

E. HYBRID MODEL EVALUATION

The evaluation of the proposed hybrid chatbot model was conducted in two main stages:

1) COMPARISON WITH INDIVIDUAL MODELS

The hybrid model was evaluated against its three components: rule-based, retrieval-based, and generative models.

Algorithm 1: MNB Classifier Algorithm

```

1: Start
2: Input:
3:    $D \leftarrow$  Training dataset
4:    $C \leftarrow$  Set of query categories
5:    $V \leftarrow$  Vocabulary
6: Output:
7:   A model that predicts the query category for new input queries
8: Preprocessing:
9:   Tokenize and preprocess the text data
10:  Build a vocabulary ( $V$ ) of all unique words in the training dataset.
11: Train the Model:
12:  Initialize:
13:    Prior probabilities  $P(c)$  for each category  $c \in C$ 
14:    Likelihood probabilities  $P(w|c)$  for each word  $w \in V$  and category  $c \in C$ 
15:  Calculate Priors:
16:    For each category  $c \in C$  do
17:       $N_c \leftarrow$  Count of queries in category  $c$ 
18:       $N \leftarrow$  Total number of queries
19:       $P(c) \leftarrow N_c/N$ 
20:    End For
21:  Calculate Likelihoods (with Laplace smoothing):
22:    For each category  $c \in C$  do
23:       $N_{c,words} \leftarrow$  Total count of all words in queries of category  $c$ 
24:      For each word  $w \in V$  do
25:         $count(w, c) \leftarrow$  Count of word  $w$  in queries of category  $c$ 
26:         $P(w|c) \leftarrow (count(w, c) + 1)/(N_{c,words} + |V|)$ 
27:      End For
28:    End For
29:  Prediction (for a new input query  $q$ ):
30:    Tokenize and preprocess  $q$  to get its words/features.
31:    For each category  $c \in C$  do
32:      Calculate the log-posterior probability:
33:       $\log P(c|q) \leftarrow \log P(c) + \sum (\log P(w|c))$ , for all words  $w \in q$ 
34:    End For
35:    Assign  $q$  to the category with the highest posterior probability:
36:    Predicted category  $\leftarrow \arg \max_c (\log P(c|q))$ 
37:  Return:
38:    Predicted category for the input query
39: End

```

FIGURE 6. Multinomial Naïve Bayes classifier algorithm.

Performance metrics, including Accuracy, Precision, Recall, and F1-Score, were used to assess each model's ability to handle structured, retrieval-based, and open-ended queries. By leveraging the strengths of all three models,

the hybrid model demonstrated superior results across all metrics, highlighting its ability to effectively handle diverse query types with improved accuracy and adaptability.

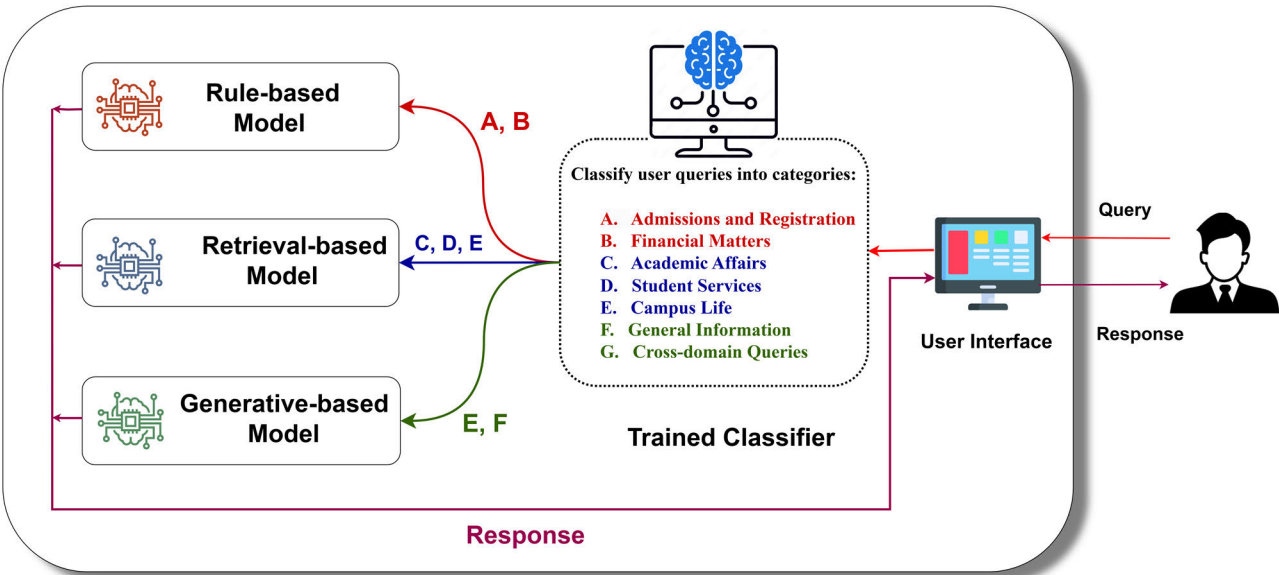


FIGURE 7. Components and operation flow of the proposed hybrid model.

Algorithm 2: Hybrid Chatbot Model Algorithm

```
1: Start
2: Input: UserQuery  $\leftarrow$  Get Query from User
3: Output: Response
4: function QUERYPROCESSINGFUNCTION(UserQuery)
5:   PredictedCategory  $\leftarrow$  Classify(UserQuery)
6:   if PredictedCategory is "A" OR "B" then
7:     Response  $\leftarrow$  RuleBasedModel(UserQuery)
8:   else if PredictedCategory is "C" OR "D" OR "E" then
9:     Response  $\leftarrow$  RetrievalBasedModel(UserQuery)
10:  else if PredictedCategory is "F" OR "G" then
11:    Response  $\leftarrow$  GenerativeBasedModel(UserQuery)
12:  else
13:    Response  $\leftarrow$  "Sorry, I cannot handle this query at the moment."
14:  end if
15:  return Response
16: end function
17: Main Program Loop:
18: Response  $\leftarrow$  QueryProcessingFunction(UserQuery)
19: Display Response to User
20: End
```

FIGURE 8. Proposed Hybrid model algorithm.

2) COMPARISON WITH BEST-PRACTICE CHATBOTS

The hybrid model was also compared to the best-practice chatbot implementations from the literature. This comparison focused on five critical factors:

- Objective/Task Specification: Ensuring alignment with diverse institutional needs.
- Data Management: Using a curated, dynamic dataset to ensure accuracy and scalability.

- **Chatbot Model Selection:** Integrating rule-based, retrieval-based, and generative approaches for enhanced adaptability.
- **Model Evaluation:** Employing a multidimensional evaluation framework for comprehensive assessment.
- **Model Accuracy:** Achieving a high accuracy rate while balancing precision and adaptability.

These comparisons demonstrated that the hybrid model not only outperformed individual models but also set a new benchmark for chatbots in educational administration by addressing critical design and evaluation factors comprehensively.

IV. RESULTS AND DISCUSSION

This section provides the results of an in-depth analysis and evaluation of the proposed hybrid chatbot model, focusing on five critical areas: Dataset Analysis, Individual Model Analysis, Models Evaluation and Comparison, Classifier Analysis, and Hybrid Model Performance. Each subsection details the development and evaluation process, demonstrating how the hybrid model integrates the strengths of individual approaches to deliver enhanced accuracy, adaptability, and user satisfaction. Additionally, comparative analysis with best-practice implementations highlights the model's scalability and effectiveness in addressing a wide range of administrative tasks, further validating its practical applicability.

A. DATASET ANALYSIS

The dataset created for this study serves as a cornerstone for validating the proposed chatbot framework, specifically tailored to support Sakarya University. Comprising 2,253 lines of conversation, the dataset was meticulously developed using diverse data sources, rigorous pre-processing techniques, and domain-specific categorization. This specialization ensures a highly relevant and organized dataset that effectively addresses the needs of Sakarya University. However, it also raises considerations regarding its scalability to other institutions.

Despite this limitation, the methodology used to construct the dataset is highly adaptable and can be replicated for other universities. By sourcing institution-specific FAQs and related resources, similar datasets can be developed to meet the unique requirements of different educational institutions. Furthermore, the modular design of the proposed hybrid chatbot model supports seamless integration with new datasets, making it a versatile solution for various educational settings. Future research could involve creating datasets for multiple institutions to evaluate the hybrid model's adaptability and robustness in diverse contexts. This would provide deeper insights into the scalability of the proposed solution and its potential to support a broader range of administrative tasks across different educational environments.

B. INDIVIDUAL MODELS ANALYSIS

Rule-based chatbots developed using AIML and the Pandorabots platform offer several advantages but also face

notable limitations. One of their primary strengths is their simplicity. AIML provides a clear and accessible framework for defining rules, making it suitable even for developers with limited programming experience. Additionally, rule-based chatbots allow for high control over responses, ensuring consistent and predictable behaviour. The Pandorabots platform further enhances the development process by offering a user-friendly interface for managing AIML files and tools for seamless integration with external systems. However, scalability poses a significant challenge for rule-based chatbots. As the number of patterns and rules grows, maintaining the knowledge base becomes increasingly complex and time-consuming. Furthermore, these chatbots often struggle with handling long or dynamic conversations, as they lack mechanisms to retain context, which can lead to a less natural and engaging user experience.

The retrieval-based chatbot demonstrated outstanding performance in delivering accurate, efficient, and contextually relevant responses during its evaluation. Utilizing advanced techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) and cosine similarity, the chatbot effectively matched user queries to the most relevant predefined responses, even when phrasing variations occurred. This capability ensured reliable handling of diverse input formats while maintaining high accuracy.

The integration of entity extraction further enhanced the chatbot's contextual understanding. By extracting specific details such as names, dates, or keywords from user inputs, the chatbot tailored responses to individual queries, providing a more personalized and meaningful interaction. Additionally, the structured JSON-based knowledge base offered a scalable and organized framework for managing domain-specific information, ensuring consistent and reliable responses across categories such as Admissions, Financial Matters, and General Information. These features collectively enabled the retrieval-based chatbot to excel in handling repetitive and standardized queries with precision and efficiency. The evaluation results validate the robustness of its architecture, highlighting its effectiveness in addressing user inquiries within the context of educational administration.

The generative-based chatbot was fine-tuned using a pre-trained Pegasus model from the Hugging Face library, leveraging a custom dataset of Q&A pairs to optimize its performance. The dataset underwent pre-processing steps, including tokenization and encoding, to ensure consistent sequence lengths across inputs. The model's weights were optimized through iterative training over 40 epochs using the Adam optimizer, which minimized loss values and progressively improved response quality. The chatbot was deployed with a graphical user interface (During evaluation), the generative-based chatbot excelled in handling open-ended and complex queries, demonstrating its ability to dynamically generate responses and adapt to ambiguous or creative inquiries. Its advanced technological foundation, combined with a user-focused design, enabled the chatbot to deliver high-quality, context-aware responses, making it

TABLE 2. Rule-based chatbot model evaluation results.

Metrics	Results %
Accuracy of Matching	210/300 = 73.33 %
Rule Coverage	191/300 = 63.33 %

TABLE 3. Retrieval-based chatbot model evaluation results.

Metrics	Results %
Response Selection Accuracy	252/300 = 83.33 %
Diversity of Responses	Qualitative assessment of the chatbot's ability to provide varied and appropriate responses is required.

TABLE 4. Generative-based chatbot model evaluation results.

Metrics	Metric	Results %
Response Quality	BLEU (0-1)	0.7812 = 78.12 %
Coherence and Relevance	Manual (1-5)	75.42 %
Grammar and Language Fluency	Manual (1-5)	87.23%

an effective and versatile tool for addressing diverse and complex applications.

C. MODELS EVALUATION AND COMPARISON

For randomly selected 300 different queries, the performance of all three chatbot approaches was evaluated. The rule-based chatbot achieved a matching accuracy of 73.33% for user queries to predefined rules, effectively generating appropriate responses. It covered 63.33% of the rules defined within AIML categories (Table 2).

On the other hand, the retrieval-based chatbot demonstrated a higher performance with a score of 83.33% in selecting the most relevant response from a predefined set of responses. Additionally, it provided a wide range of response diversity, enhancing the overall user experience (Table 3).

To evaluate the generative-based chatbot, the chatbot's responses for the same set of queries were assessed by an English specialist annotator. The manual evaluation using the Likert scale (strongly agree(5), agree(4), neutral(3), disagree(2), strongly disagree(1)) revealed that the model achieved scores of 72.12%, 75.42%, 87.23% for response quality, coherence, grammar, and language fluency, respectively (Table 4).

The evaluation results underscored the need for enhancements across all three chatbot models and the dataset. For the rule-based model, expanding the AIML categories was essential to address a broader range of user queries. Consequently, an additional 112 categories were added, significantly improving the model's query coverage.

For the retrieval-based chatbot, the random() function was activated to introduce greater response diversity, particularly for queries with multiple valid answers. This adjustment aimed to enhance the user experience by offering varied and dynamic responses, making interactions more engaging.

To improve the performance of the generative-based model, the dataset was augmented with additional examples designed to better account for comprehension and context. This enhancement ensured the model generated more contextually appropriate and coherent responses, particularly for complex and ambiguous queries.

The evaluation results are detailed in Table 5, which lists the questions included in the user questionnaire, and Table 6, which summarizes the models' performance based on user interactions. Scores for each question, along with the overall performance of the models, were calculated using Equation 9.

- User Satisfaction: Derived from responses to questions 2, 6, and 7, the retrieval-based model achieved the highest score in this category, reflecting its effectiveness in providing relevant and engaging responses.
- Model Usefulness: Assessed through questions 3, 8, and 9, the rule-based model performed best, demonstrating its reliability in addressing structured and straightforward queries.
- User Engagement: Evaluated using questions 5 and 10, the rule-based model also excelled in this area, showcasing its ability to maintain user interest during interactions.
- Human-like Conversation: Question 4 measured the models' ability to simulate human-like dialogue, with the generative-based model receiving the top score due to its dynamic and context-aware response generation.

These findings provided valuable insights for iterative improvements, highlighting the strengths of each model and guiding efforts to refine the chatbot framework for enhanced adaptability and user satisfaction.

Score =
$$\frac{(Summation\ of\ all\ 267\ students'\ responses)}{10 * 267} \times 2 \tag{9}$$

The equation normalizes and scales user responses from the questionnaire, originally rated on a 1–5 scale, into a 0–10 range to enable consistent comparison across chatbot models. It aggregates scores collected from 267 students, divided by the maximum possible score (adjusted for the 10-point scale), and multiplies by 2 to align with the desired range. This normalization ensures that the calculated scores effectively represent key metrics such as user satisfaction, usefulness, engagement, and conversational ability. By standardizing the results, the scores become easier to interpret and facilitate direct comparisons among the performance of the three chatbot models.

The evaluation results and user feedback highlighted distinct strengths for each chatbot model. The rule-based model excelled in model usefulness and user engagement, primarily

TABLE 5. User survey questions with rating scales.

No.	Questions	Rating scales
Q1	How would you rate the accuracy of the responses?	(1-5)
Q2	How would you rate your level of acceptance of the chatbot's suggestions or recommendations?	(1-5)
Q3	How would you rate the usefulness of the chatbot in addressing your queries or providing information?	(1-5)
Q4	How would you evaluate the context of the conversations?	(1-5)
Q5	Would you be interested in using the chatbot again in the future?	(1-5)
Q6	How satisfied are you with the chatbot's ability to understand your queries?	(1-5)
Q7	How satisfied are you with the chatbot's response time?	(1-5)
Q8	How would you rate the chatbot's ability to provide relevant and helpful information?	(1-5)
Q9	How would you rate the chatbot's overall performance in assisting you with your needs?	(1-5)
Q10	How likely are you to recommend the chatbot to others?	(1-5)

TABLE 6. Conducted user surveys results.

User Experience	Rule-based (out of 10)	Retrieval-Based (out of 10)	Generative-based (out of 10)
User Satisfaction	7.29	7.53	6.78
Usefulness/Effectiveness	7.66	7.11	7.45
User Engagement	8.33	7.56	8.10
Human-like Conversation	5.79	6.22	7.24

due to its ability to deliver precise and accurate answers for specific categories such as tuition fees and course schedules. Its user interface, enhanced with rich media tags available in the AIML playground (Pandorabots), further contributed to its appeal and ease of use. The retrieval-based model stood out in response selection, achieving the highest user satisfaction scores. This performance was attributed to its robust ability to match user queries with the most relevant predefined responses, ensuring both accuracy and relevance. The generative-based model demonstrated exceptional performance in providing a more human-like conversational experience, leveraging its implementation of a generative pre-trained model with a transformer architecture. This capability allowed it to handle open-ended and dynamic interactions more effectively.

To evaluate the models' strengths in handling various administrative tasks at Sakarya University and to inform the

TABLE 7. Task completion rate for all three models.

Tasks	Query's Category	Rule-based	Retrieval-based	Generative-based
Task1	Admissions and Registrations	87.33	82.13%	81.21%
Task2	Academic Affairs	78.18%	84.67%	77.57%
Task3	Financial Matters	89.56%	82.33%	79.77%
Task4	Student Services	55.51%	71.18%	78.67%
Task5	Campus Life	84.35%	91.44%	83.09%
Task6	Greetings and General Information	82.57%	76.13%	97.89%
Task7	Cross-domain Queries	44.21%	51.16%	73.12

design of the proposed hybrid model, a detailed analysis was conducted on all recorded conversations. Table 7 summarizes the task completion rates for each model:

- The rule-based model performed particularly well in the categories of Admissions and Registrations (87.33%) and Financial Matters (89.56%), demonstrating its effectiveness in structured queries.
- The retrieval-based model achieved the highest results in Academic Affairs (84.67%), Student Services (88.94%), and Campus Life (91.44%), showcasing its ability to handle semi-structured queries with precision.
- The generative-based model excelled in the categories of General Information (100%) and Cross-domain Queries (73.12%), underscoring its adaptability and dynamic response capabilities.

These findings emphasize the complementary strengths of the three models, validating the potential of a hybrid approach to leverage their advantages for enhanced versatility and performance.

The categorization of chatbot performance stemmed from the study's analysis, where factors such as the specific implementation of each chatbot model, the quality of the training data, and the training methodology influenced the distinct strengths exhibited by each model across various task categories. During the evaluation process, individual chatbot responses were thoroughly analysed to assess their performance. As detailed in Table 8, key performance metrics such as Accuracy, Precision, Recall, and F-Measure were computed using standard formulas [53]. These metrics provided a quantitative assessment of each model's ability to handle queries effectively, offering insights into their strengths and limitations. This detailed evaluation played a crucial role in understanding how the combination of model architecture, data quality, and training strategies contributed to the overall performance of the chatbot system.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

where:

TABLE 8. Accuracy, precision, recall, and F1-measure for all three models.

Response types	Rule-based	Retrieval-Based	Generative-based
True Positive	302	356	323
False Positive	132	98	111
False Negative	73	47	57
True Negative	11	17	25
Accuracy	76.11%	87.64%	83.78 %
Precision	0.70	0.78	0.74
Recall	0.81	0.88	0.85
F1-Measure	0.62	0.67	0.65

True Positive (TP): Refers to the number of instances where the chatbot correctly gives a positive response.

False Positive (FP): Refers to the number of instances where the chatbot incorrectly gives a positive response.

True Negative (TN): Refers to the number of instances where the chatbot correctly gives a negative response.

False Negative (FN): Refers to the number of instances where the chatbot incorrectly gives a negative response.

Accuracy measures the overall correctness of the chatbot in predicting user intents or matching queries with the correct response categories.

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

Precision evaluates the proportion of correctly matched user queries (true positives) out of all predicted matches (true positives + false positives). For the chatbot, it reflects how well it avoids giving irrelevant responses to user queries.

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

Recall measures the chatbot's ability to correctly match all relevant user queries. It indicates how often the chatbot successfully identifies the correct intent or response.

$$F1 - score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (13)$$

The F1-Score provides a balanced measure between Precision and Recall, offering insight into the chatbot's effectiveness in handling user queries with accuracy and relevance.

Accuracy assesses the overall correctness of the model's predictions, with the retrieval-based model achieving the highest accuracy among the three chatbot models. However, relying solely on accuracy does not provide a comprehensive evaluation of performance. To gain deeper insights into the models' capabilities, additional metrics such as precision, recall, and F1-Measure were analyzed.

- Precision, which evaluates the proportion of correctly predicted positive responses out of all predicted positives, showed that the retrieval-based model performed

best, achieving a score of 78%. This indicates its ability to provide accurate responses in the majority of cases.

- Recall, which measures the model's ability to identify all actual positive responses, also favored the retrieval-based model, with a success rate of 88%. This highlights its effectiveness in capturing relevant responses.
- The F1-Measure, the harmonic mean of precision and recall, further emphasized the retrieval-based model's balanced performance, demonstrating its strength in handling user queries effectively.

The evaluation process aimed to identify the strengths and weaknesses of each model, providing valuable insights to guide improvements in both the dataset and model design. These findings also formed the basis for developing a hybrid system that integrates the unique strengths of all three chatbot models. Based on the evaluation results, necessary adjustments were made to the dataset and model architecture to optimize the system and support the initialization of the classifier, ensuring enhanced overall performance.

D. CLASSIFIER ANALYSIS

The evaluation and comparison of the models revealed distinct strengths in addressing different categories of queries. The rule-based model effectively handled queries related to Admissions and Registrations and Financial Matters, leveraging its structured framework for delivering precise responses. The retrieval-based chatbot excelled in managing queries concerning Academic Affairs, Student Services, and Campus Life, utilizing its ability to match user inputs to relevant knowledge base entries. Meanwhile, the generative-based model demonstrated superior performance in addressing queries categorized under General Information and Cross-domain, showcasing its adaptability and capability to handle ambiguous or open-ended inquiries effectively. These findings informed the development of the dataset described in Section III-D, which was used to train and evaluate the classifiers. The performance of the classifiers was rigorously assessed using a testing set, employing key metrics such as accuracy, precision, recall, and F1-score:

- Accuracy measured the proportion of correctly classified queries.
- Precision evaluated the relevance of the classifications.
- Recall assessed the proportion of actual queries retrieved correctly.
- F1-score provided a balanced measure combining precision and recall.

The evaluation demonstrated that the Multinomial Naive Bayes (MNB) classifier delivered robust performance across all output labels, as highlighted in Table 9. Among the classifiers:

- The Neural Network classifier achieved the highest accuracy of 97.12% and excelled in precision, recall, and F1-score metrics. However, its high computational demands, including a training time of 49.81 seconds and a prediction time of 3.58 ms per query, limited

TABLE 9. Accuracy, precision, recall, and F1-measure, training time, prediction time for the classifiers.

Classifier	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Training Time (s)	Prediction Time (ms/query)
Multinomial NB	96.31	95.75	96.12	96.05	1.12	0.63
SVM (Linear)	92.36	91.59	91.82	91.70	14.72	1.41
Random Forest	91.68	91.42	91.65	91.53	8.34	2.21
Neural Network	97.12	96.51	96.72	96.61	49.81	3.58

TABLE 10. Accuracy, precision, recall, F1-measure for hybrid model.

Tasks	Hybrid model
True Positive	477
False Positive	7
False Negative	13
True Negative	21
Accuracy	96.13%
Precision	0.9855
Recall	0.9734
F1-Measure	0.9749

its feasibility for real-time applications in resource-constrained environments.

- The MNB classifier, in contrast, achieved an accuracy of 96.31%, closely rivalling the Neural Network while maintaining exceptional computational efficiency. With a training time of just 1.12 seconds and a prediction time of 0.63 ms per query, MNB emerged as a practical choice for real-time deployment in educational administrative systems.
- SVM and Random Forest classifiers achieved moderate accuracy scores of 92.36% and 91.68%, respectively, but required more computational resources than MNB.

The MNB classifier was employed to categorize queries into predefined labels, which were carefully structured to ensure comprehensive coverage of student inquiries. These labels struck a balance between specificity and flexibility:

- Specific labels targeted well-defined administrative tasks, such as Admissions and Registrations, Financial Matters, Academic Affairs, Student Services, and Campus Life.
- General labels, such as Greetings, General Information, and Cross-domain Queries, were designed to handle less structured or ambiguous queries, enhancing the system's adaptability.

This two-tiered labelling structure enabled the chatbot to manage focused queries with precision while effectively addressing broader, less-defined interactions. By integrating

the MNB classifier into the hybrid chatbot model, the system achieved scalability, adaptability, and high performance. The classifier efficiently routed queries to the appropriate model: rule-based, retrieval-based, or generative ensuring that both simple and complex queries were handled with accuracy and contextual relevance. This design makes the hybrid chatbot model well-suited to meet the diverse administrative requirements of educational institutions.

E. PERFORMANCE OF THE HYBRID MODEL

This subsection evaluates the performance of the proposed hybrid chatbot model, beginning with a comparison to the individual models: rule-based, retrieval-based, and generative to highlight its superior accuracy and adaptability, as shown in Table 10. By leveraging the strengths of all three approaches, the hybrid model achieves significant advancements in handling a diverse range of queries. The evaluation then examines the hybrid model's performance against existing best-practice chatbot implementations, as presented in Table 11, to demonstrate its comprehensive and robust design.

Table 10 highlights the notable improvement in accuracy achieved by the hybrid model when using the updated dataset, with an equal number of queries employed for all three models. These results underscore the hybrid model's ability to effectively integrate the strengths of rule-based systems for structured queries, retrieval-based approaches for precise response matching, and generative models for handling complex and open-ended inquiries. This achievement marks a significant contribution to the development of chatbots for educational administration, showcasing the hybrid model's ability to address a wide array of user needs with improved accuracy and adaptability.

The development and evaluation of chatbot systems for educational administration rely on addressing five critical factors: Objective/Task Specification, Data Management, Chatbot Model Selection, Model Evaluation, and Model Accuracy. These factors significantly shape the chatbot's design, functionality, and overall effectiveness. While existing studies have made advancements in some areas, many fail to address these factors comprehensively, limiting their applicability and performance. Below is a detailed discussion

TABLE 11. A summary of comparison between the proposed hybrid model and related works.

Reference	Objective	Dataset used	Dataset Size	Model type	Evaluation Approach	Performance
[3]	Addressing queries about admission for new students, tuition fees, IELTS test, scholarships, or deadlines.	National Economics University dataset	1500 examples	Retrieval-based	F1-score, accuracy, and precision	95.1%
[55]	Answering queries related to postgraduate studies at Princess Nourah University	Frequently Asked Questions (FAQs)	-	Rule-based and Retrieval-based approaches	Chatbot Usability Questionnaire (CUQ) and Accuracy	91.97%
[45]	Intelligent customer service system for university admissions	encyclopedia question-and-answer, Tamkang University Dataset	1.5 million pre-filtered questions and answers, 210 questions and answers about Tamkang University	Generative-based	BLEU, questionnaire-based evaluation	80.02%
[57]	Provide information to users about various campus life	Mississippi State University Dataset	-	Retrieval-based approach	subjective satisfaction surveys using the System Usability Scale (SUS)	96%
[59]	Provide information regarding regulations, admissions, scholarships, and social clubs	Baskent University Dataset	688 question-answer pairs.	Generative-based approach	BLEU And METEOR, ROGUE-1 metrics	81%
[36]	Addressing FAQ, student support	National University of Science & Technology (NUST), Islamabad, Pakistan dataset	1000 examples	Retrieval-based	Precision, fl-score, and accuracy	76.8%
[56]	Assistance providing to technical and vocational students in administrative and academic queries	College related dataset.	265 examples	Retrieval-based approach	Precision, Recall, F-1 measure	90%
[19]	Answer queries regarding registration procedures and tuition payments	student queries and frequently asked questions	60 student queries and frequently asked questions	Rule-based and retrieval-based methods	quantitative measures of ease of use and user satisfaction	The model achieved above 3 out of 5 in most evaluation cases
Proposed Hybrid Model	Proposing a hybrid educational administrative chatbot	Sakarya University Dataset	2253 Question-Answer pairs	Hybrid approach	Quantitative (Accuracy, Precision, Recall, F-1 measure) and Qualitative (Survey and User Feedback)	97.57%

of each factor, supported by relevant literature, along with a comparison of the proposed hybrid chatbot model to best-practice studies (Table 11) to highlight its superior performance.

1. Objective/Task Specification

Clearly defining the administrative tasks a chatbot is expected to handle is a foundational step in its development. Research emphasizes that aligning a chatbot's objectives with an institution's unique requirements ensures relevance and effectiveness in addressing specific challenges [1]. For

instance, the chatbot developed for the National Economics University focuses on tasks like admission queries, tuition fees, IELTS tests, and scholarships [3]. While effective within these domains, it struggles with broader administrative tasks. In contrast, the proposed hybrid chatbot model dynamically tailors responses across multiple query types, offering alignment with institutional goals and flexibility to handle diverse administrative needs. This ensures the system's applicability to both specific and generalized scenarios, enhancing its overall utility.

2. Data Management

The quality and management of data are crucial to the chatbot's reliability and user satisfaction. Effective data management requires curating datasets that adhere to privacy, security, and ethical standards [42], [54]. For instance, the chatbot at Princess Nourah University uses a static FAQ-based dataset, limiting its adaptability to evolving user needs [55]. Similarly, the dataset used in the study [56], with only 265 samples, lacks the diversity needed to handle complex or varied queries, reducing its robustness and scalability. By leveraging a well-curated and dynamically updated dataset, the hybrid chatbot model ensures consistent accuracy and relevance. This dynamic approach addresses gaps in data management observed in other implementations, ensuring the chatbot remains responsive to diverse user needs.

3. Chatbot Model Selection

Selecting the appropriate response model: whether rule-based, retrieval-based, or generative is integral to a chatbot's functionality. Hybrid models, which adapt dynamically to conversational contexts, are shown to balance efficiency and adaptability, making them ideal for complex tasks in educational administration [44], [45]. For instance, the chatbot developed for Ho Chi Minh City University employs a retrieval-based approach to support administrative and learning inquiries [57]. However, its reliance on predefined data limits its ability to handle open-ended queries effectively. The proposed hybrid model integrates rule-based, retrieval-based, and generative components, enabling it to address a wider range of query types. This multi-faceted approach allows the system to maintain precision in structured queries while offering adaptability for complex, open-ended tasks.

4. Model Evaluation

Comprehensive evaluation ensures that the chatbot meets its objectives and provides a positive user experience. Regular assessment using user feedback, automated methods, and data analysis is essential for maintaining relevance and addressing evolving requirements [58]. For instance, the chatbot developed for Telkom University primarily uses BLEU scores to evaluate generative responses [59]. While effective for assessing linguistic accuracy, such evaluations often overlook user satisfaction or task-specific performance. The hybrid model employs a multidimensional evaluation framework that incorporates metrics such as accuracy, user satisfaction, response time, and contextual adaptability. This holistic approach ensures superior performance and highlights areas

for further improvement, resulting in a more robust and user-friendly system.

5. Model Accuracy

Accuracy remains a cornerstone metric for chatbot performance. However, the literature advocates for a holistic approach that integrates multiple metrics to capture a model's true capabilities and identify improvement areas [58]. For example, the chatbot developed for NUST achieves a 76.8% accuracy rate, primarily focusing on FAQ-style queries [36]. While satisfactory for limited tasks, it lacks the flexibility to adapt to diverse query contexts. In contrast, the hybrid chatbot model achieves a high accuracy of 97.57%, excelling in adaptability and contextual understanding. By balancing precision and recall, the model delivers accurate and contextually relevant responses across diverse administrative tasks, demonstrating a comprehensive approach to evaluation.

By addressing these factors comprehensively, the proposed hybrid chatbot model sets a new benchmark for chatbots in educational administration, as highlighted in Table 11. Its superior performance, adaptability, and scalability underscore the importance of incorporating these critical considerations into chatbot development. The hybrid framework optimizes administrative efficiency and lays the foundation for extending its application to other domains, such as healthcare and e-commerce, showcasing its potential for cross-domain scalability and robust performance.

V. CONCLUSION AND FUTURE WORK

A. CONCLUSION

This study explored the potential of a hybrid chatbot model that combines rule-based, retrieval-based, and generative-based chatbot models with the aim of enhancing educational administration. The proposed hybrid model employs a query routing classifier to leverage the strengths of each of the three integrated models. The hybrid model evaluation result shows a significant accuracy improvement of 97.57%, outperforming individual models. The hybrid model demonstrated its ability to optimize both efficiency and user experience within educational settings. Comparative analysis with state-of-the-art solutions further highlighted its competitive performance, adaptability, and capacity to address diverse administrative needs. The study's findings demonstrate the flexibility and scalability of this approach, making it a robust solution for addressing the evolving needs of educational administration and other domains.

B. FUTURE WORK

To extend the applicability and effectiveness of the hybrid chatbot model, future work should focus on:

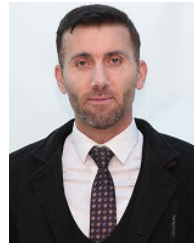
1. Expanding the Dataset: Enriching the dataset with more diverse queries to enhance the model's robustness and ability to address a broader range of inquiries.
2. Exploring Advanced Classification Algorithms: Investigating deep learning-based classification techniques to improve query routing accuracy and overall system performance.

3. IoT-Driven Data Collection and Integration: Incorporating IoT technologies for real-time, context-aware responses. For example, integrating data for personalized notifications or resource management in smart campus environments.
4. Adapting the Model for Other Domains: Modifying the dataset to adapt the hybrid model for other sectors, such as healthcare and e-commerce.

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